OPERATIONS RESEARCH · PROJECT REVIEW 1

Dynamic EMS Dispatch & Station Placement Optimization

Optimizing where ambulances wait and how they are dispatched, using real Seattle 911 data

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WHY THIS PROJECT

The Problem & Its Significance

In a medical emergency, survival is decided in minutes. Two operational decisions control how fast an ambulance actually reaches a patient — and both can be optimized.

Q1

Where to wait?

Placing stations well means more of the city sits within a short response-time radius.

Q2

Who to send?

During simultaneous emergencies, smart ambulance-to-incident matching cuts everyone's wait.

Why it matters

Emergency response is one of the oldest and most studied applications of operations research and a domain where better models translate directly into lives.

We apply it to a real city with real data.

WHAT WE WILL DELIVER

Project Objectives

Build an interactive application that turns real emergency data into optimized decisions:

1

Interactive data exploration

Load a chosen period of Seattle EMS data and generate statistics on demand — busiest zones, peak hours, demand patterns.

2

Optimal station placement

Use the covering model to determine where ambulance stations should be located across the city.

3

Optimal dispatch

Use the assignment model to match available ambulances to incidents and minimize total response time.

4

Visual decision support

Present the statistics and optimization results on an interactive city map.

THE DATA

Our Dataset: Seattle 911 Calls

~2.2M	Every 5 min	Public Domain
incident records	update frequency	open license

911

Seattle Real-Time Fire 911 Calls

Source: City of Seattle Open Data Portal (data.seattle.gov) — dataset ID kzjm-xkqj.

The DEMAND side of our project: it records where and when real emergencies occur across the city.

THE DATA

What's Inside the 911 Dataset

Field	Meaning	Used for
address / latitude / longitude	Exact incident location	Mapping demand, distances
type	Kind of call (medical, aid, fire…)	Filtering to EMS calls only
datetime	Date and time of the call	Peak-hour stats, dispatch scenario
zip code (computed)	Which zone the incident is in	Grouping demand into zones
incident number	Unique ID per call	De-duplication

WHAT WE ASK

The Questions We Optimize

Covering Model

- Minimum stations to cover the whole city within an X-minute limit?
- With only N stations, where do they maximize covered demand?
- How does coverage change at 5 vs 8 vs 10 minutes?
- How does our placement compare to Seattle's real stations?

AHP

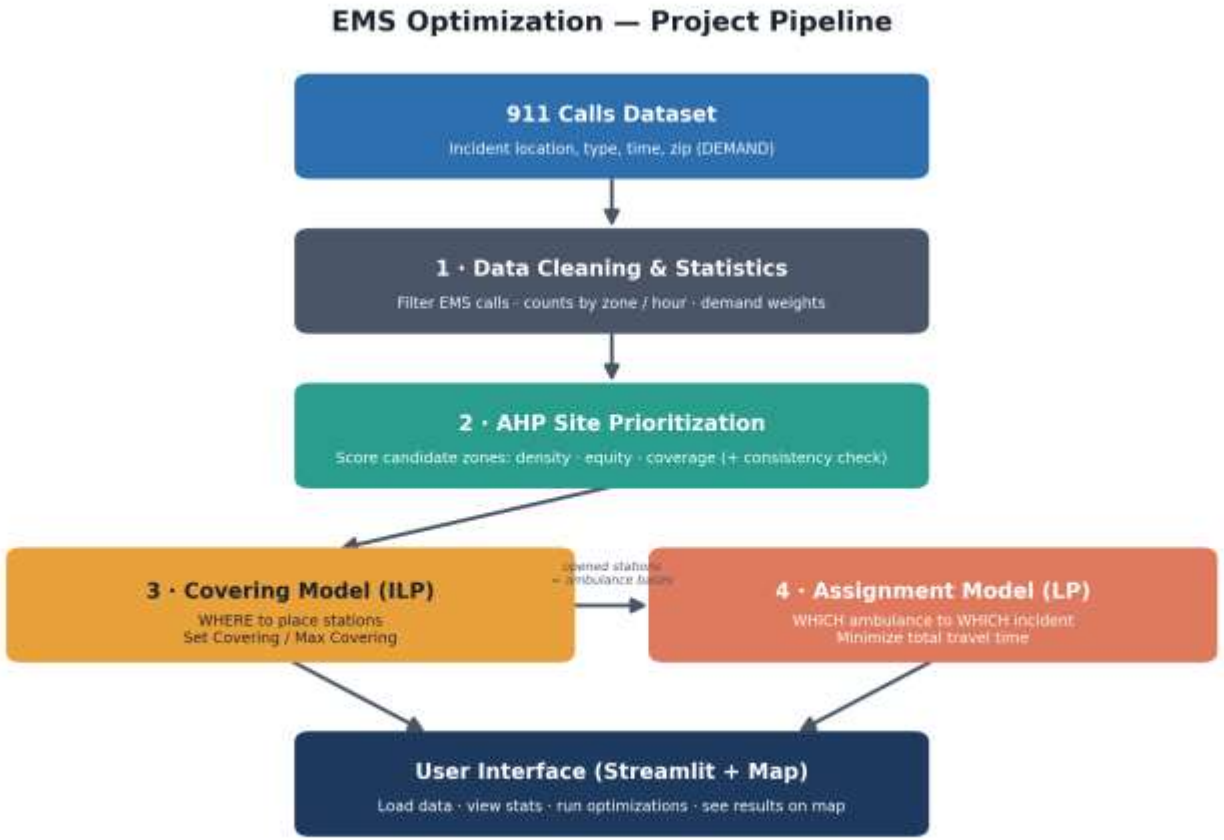
- Which candidate areas rank highest when we weigh demand density, distance from existing help, and coverage potential together?

Assignment Model

- In a peak hour of simultaneous emergencies, which ambulance goes to which incident to minimize total response time?

METHODOLOGY

How Everything Connects



Data in → clean & compute stats → AHP ranks sites → two optimizations run → results on the map

METHODOLOGY · STEP 1

Data Cleaning & Statistics

We filter the data to genuine EMS/aid calls, group incidents by zip-code zone, and compute statistics the user can view on demand. These counts also become each zone's demand weight for the models.

Busiest zones

Incident counts per zone — where demand concentrates.

Peak timing

Calls by hour and weekday — when load spikes.

Demand weights

Per-zone importance fed into both optimizations.

METHODOLOGY · STEP 3

Optimization 1 — Covering Model (ILP)

Decide WHERE to place stations. A zone is 'covered' if a station can reach it within a user-set response-time limit. Two modes:

Mode A — Set Covering

Determines the minimum number of stations required such that every demand zone is served within the specified response time threshold. Each candidate location is a binary decision (open or not), and the model guarantees complete coverage of the city at the lowest possible facility count.

Objective: *Minimize number of stations*

Mode B — Maximal Covering

Fixes the number of stations at a user-defined limit and positions them to serve the greatest possible share of demand within the response-time threshold. This reflects real-world budget constraints, where full coverage may be infeasible and coverage must be maximized under a fixed number of facilities.

Objective: *Maximize demand covered*

METHODOLOGY · STEP 4

Optimization 2 — Assignment Model (LP)

Decide WHICH ambulance goes to WHICH incident during a busy hour, so that total travel time across all responders is minimized.

The setup

Incidents = real calls from a chosen peak hour

Ambulances = one ready unit stationed at each base

Each incident → exactly one ambulance

Each ambulance → at most one incident

Objective: minimize total travel time

The connecting idea

The station locations determined by the covering model serve as the ambulance bases in the assignment model. This linkage integrates both optimizations into a single, coherent framework — the core contribution of our project.

(Unequal counts handled with dummy rows/columns — the transportation-problem link.)

METHODOLOGY · STEP 2

AHP — Multi-Criteria Site Scoring

AHP combines several competing criteria into one priority score per candidate area, so we can rank which zones most deserve a station.

THE THREE CRITERIA

Incident Density

Number of emergencies in the area.

Equity Gap

Distance to the nearest existing station (farther = higher need).

Coverage Potential

How many zones a station here could reach quickly.

BUILDING THE WEIGHTS

1 · Pairwise comparison

Compare the criteria two at a time on the Saaty 1–9 scale (1 = equally important … 9 = extremely more important). For 3 criteria, just 3 comparisons.

2 · Derive the weights

The comparison matrix is converted into a weight for each criterion (all summing to 100%), capturing how much each factor matters.

Consistency Check — keeping the judgement logical

Because the pairwise ratings are human judgements, they can quietly contradict each other. AHP measures this: from the matrix it computes **λmax** → a Consistency Index (CI) → a **Consistency Ratio (CR)**. Rule: **CR < 0.1** means the judgements are consistent and the weights are trustworthy; if higher, we revise the comparisons. Each member fills the table independently and we average the results.

HOW WE BUILD IT

Tools & Tech Stack

TOOL

Python + pandas

Loading data & computing statistics

TOOL

PuLP

Solving the covering (ILP) & assignment (LP) models

TOOL

Streamlit

The interactive user interface

TOOL

Folium / Plotly

Mapping incidents & stations

TOOL

Haversine

Estimating travel distance / time

TOOL

Data ingestion

Export CSV from portal / API into the app

DYNAMIC EMS DISPATCH & STATION PLACEMENT OPTIMIZATION

Thank You